

Introductory Statistics for Econometrics

Unit 3. Joint, Marginal and Conditional Distributions

Two random variables: joint p.m.f./p.d.f., marginals, independence, and conditional distributions

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Learning outcomes. After this unit, the student should be able to (i) describe the joint distribution of two random variables via p.m.f., p.d.f. or c.d.f.; (ii) compute marginal distributions from a joint distribution; (iii) test independence using the factorisation criterion; (iv) construct conditional distributions $f_{Y|X}(y | x)$; (v) reason about the support of bivariate distributions.

Source material: condensed from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 6–7 (with a preview of Lecture 13).

1 Why two variables?

Most economic questions involve relations between variables, not the behaviour of one variable in isolation:

- *Twins' IQs.* The IQ of one twin and the IQ of the other are clearly related — but how strongly?
- *Schooling and earnings.* Are years of education X and hourly wages Y statistically dependent? By how much?
- *Survival analysis.* Time X to a relapse and time Y to death are jointly determined by the underlying disease process.

To answer such questions we need the *joint distribution* of (X, Y) , which encodes everything a probabilist can know about the pair. This unit develops that machinery, paving the way for covariance, correlation and the conditional expectation function in Unit 6.

2 Joint p.m.f. and joint p.d.f.

Let (X, Y) be a pair of random variables defined on a common sample space.

Definition 2.1 (Joint p.m.f.). If both X and Y are discrete with values $\{x_i\}$ and $\{y_j\}$, the *joint probability mass function* is

$$f_{XY}(x, y) = \mathbb{P}(X = x, Y = y).$$

It satisfies $f_{XY} \geq 0$, $\sum_{i,j} f_{XY}(x_i, y_j) = 1$, and for any $A \subseteq \mathbb{R}^2$, $\mathbb{P}((X, Y) \in A) = \sum_{(x,y) \in A} f_{XY}(x, y)$.

Definition 2.2 (Joint p.d.f.). If (X, Y) is continuous, a function $f_{XY} : \mathbb{R}^2 \rightarrow [0, \infty)$ is a *joint probability density function* if for every $A \subseteq \mathbb{R}^2$,

$$\mathbb{P}((X, Y) \in A) = \iint_A f_{XY}(x, y) \, dx \, dy.$$

A joint p.d.f. must satisfy $\iint_{\mathbb{R}^2} f_{XY}(x, y) dx dy = 1$. As in the univariate case, $\mathbb{P}((X, Y) = (x, y)) = 0$ for any single point in the continuous case, and any one-dimensional curve in the plane has probability zero.

Example 2.3 (Discrete joint distribution: contingency table). Let X be the number of customers in the regular checkout line of a supermarket and Y the number in the express line. The joint p.m.f. is displayed in the body of the table below; the right margin and bottom row will be the marginal distributions discussed in Unit 3.

$f_{XY}(x, y)$	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$y = 4$	$f_X(x)$
$x = 0$	0.10	0.05	0.05	0	0	0.20
$x = 1$	0.05	0.20	0.20	0.05	0	0.50
$x = 2$	0	0	0.10	0.10	0.05	0.25
$x = 3$	0	0	0	0	0.05	0.05
$f_Y(y)$	0.15	0.25	0.35	0.15	0.10	1.00

Example 2.4 (Bivariate uniform / geometric probability). Suppose (X, Y) is uniformly distributed over a rectangle of area S . Then $f_{XY}(x, y) = 1/S$ on the rectangle and 0 outside, so for any region A inside the rectangle, $\mathbb{P}((X, Y) \in A) = \text{Area}(A)/S$. Computing probabilities reduces to a geometric exercise.

Example 2.5 (Independent exponentials). Two spark plugs in a lawn mower have lifetimes $X, Y \geq 0$ with

$$f_{XY}(x, y) = \lambda^2 e^{-\lambda(x+y)} \mathbf{1}\{x \geq 0, y \geq 0\}.$$

The probability that both plugs fail within T hours is

$$\mathbb{P}(X \leq T, Y \leq T) = \int_0^T \int_0^T \lambda^2 e^{-\lambda(x+y)} dy dx = (1 - e^{-\lambda T})^2.$$

as T increase the probability of both plugs failing converges to 1.

Joint c.d.f.

Definition 2.6 (Joint c.d.f.). The *joint cumulative distribution function* is

$$F_{XY}(x, y) = \mathbb{P}(X \leq x, Y \leq y), \quad (x, y) \in \mathbb{R}^2.$$

For continuous (X, Y) , $F_{XY}(x, y) = \int_{-\infty}^y \int_{-\infty}^x f_{XY}(u, v) du dv$ and $f_{XY}(x, y) = \partial^2 F_{XY} / \partial x \partial y$. Probabilities of rectangles are recovered by inclusion–exclusion:

$$\mathbb{P}(a < X \leq b, c < Y \leq d) = F_{XY}(b, d) - F_{XY}(a, d) - F_{XY}(b, c) + F_{XY}(a, c). \quad (1)$$

3 Marginal distributions

If we know the joint distribution of (X, Y) , the distribution of X alone is obtained by *summing or integrating out* the other variable.

Proposition 3.1 (Marginals).

- *Discrete*: $f_X(x) = \sum_y f_{XY}(x, y)$, $f_Y(y) = \sum_x f_{XY}(x, y)$.
- *Continuous*: $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$, $f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$.

The right column of the table in Example 2.3 is the marginal f_X obtained by summing across rows; the bottom row is the marginal f_Y . For Example 2.5,

$$f_X(x) = \int_0^\infty \lambda^2 e^{-\lambda(x+y)} dy = \lambda e^{-\lambda x}, \quad f_Y(y) = \lambda e^{-\lambda y}.$$

Important. The marginals *do not* in general determine the joint distribution: many different joints can share the same marginals. This is precisely why dependence is interesting — it is information *beyond* the marginals.

4 Independence of random variables

Definition 4.1 (Independence). Random variables X and Y are *independent*, written $X \perp Y$, if for any sets $A, B \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B).$$

The definition is hard to verify directly. The following criterion makes it operational.

Theorem 4.2 (Factorisation criterion). X and Y are independent if and only if their joint and marginal p.m.f. (or p.d.f.) satisfy

$$f_{XY}(x, y) = f_X(x) f_Y(y) \quad \text{for all } (x, y). \quad (2)$$

Equivalently, $F_{XY}(x, y) = F_X(x) F_Y(y)$ for all (x, y) .

Remark 4.3 (A practical shortcut). For continuous distributions, if there exist functions g and h — not necessarily proper densities — such that $f_{XY}(x, y) = g(x) h(y)$ on a *rectangular* support $S_X \times S_Y$ and zero elsewhere, then $X \perp Y$. The marginals are then $f_X(x) \propto g(x)$ and $f_Y(y) \propto h(y)$, with normalising constants absorbed.

Example 4.4 (Spark plugs are independent). In Example 2.5, $f_{XY}(x, y) = \lambda^2 e^{-\lambda x} e^{-\lambda y} = f_X(x) f_Y(y)$ on the rectangle $\{x \geq 0\} \times \{y \geq 0\}$. Hence $X \perp Y$.

Example 4.5 (Non-rectangular support breaks independence). Let $f_{XY}(x, y) = c x^2 y$ for $x^2 \leq y \leq 1$, zero otherwise. Although the algebraic expression looks separable, the support is not a rectangle: the admissible y depends on x . Concretely

$$\mathbb{P}(X \geq \tfrac{1}{2}, Y \leq \tfrac{1}{4}) = 0 < \mathbb{P}(X \geq \tfrac{1}{2}) \mathbb{P}(Y \leq \tfrac{1}{4}),$$

so $X \not\perp Y$.

Two rules of thumb.

- (i) Independence requires a rectangular (possibly all of $\mathbb{R}^2$) *support*. Non-rectangular support ruins independence automatically.
- (ii) Within a rectangular support, separability of f_{XY} is necessary and sufficient.

Example 4.6 (Constructing a table under independence). Marriage-quality $X \in \{1, 2, 3\}$ has marginal (0.291, 0.323, 0.386) and years-married $Y \in \{1, 8, 12\}$ has marginal (0.233, 0.428, 0.339). Under independence $\tilde{f}_{XY}(x, y) = f_X(x) f_Y(y)$:

$\tilde{f}_{XY}$	$y = 1$	$y = 8$	$y = 12$	f_X
$x = 1$	6.78%	12.45%	9.88%	29.12%
$x = 2$	7.52%	14.81%	10.96%	32.28%
$x = 3$	8.99%	16.50%	13.10%	38.60%
f_Y	23.30%	42.76%	33.94%	100.00%

The actual data show $\tilde{f}_{XY}$ over-states the off-diagonal cells and under-states the diagonal — a sign of dependence between marriage quality and tenure. Whether the discrepancy is statistically significant will be the subject of hypothesis testing in Unit 10.

5 Conditional distributions

When X and Y are dependent, knowing the value of X changes the distribution of Y . This is captured by the conditional distribution.

Definition 5.1 (Conditional p.m.f./p.d.f.). For $f_X(x) > 0$, the *conditional distribution* of Y given $X = x$ is

$$f_{Y|X}(y | x) = \frac{f_{XY}(x, y)}{f_X(x)}. \quad (3)$$

For each fixed x , $y \mapsto f_{Y|X}(y | x)$ is a valid one-variable p.m.f. (or p.d.f.), so it integrates / sums to one over y .

Proposition 5.2 (Equivalent characterisations). *The following are equivalent:*

- (i) X and Y are independent;
- (ii) $f_{Y|X}(y | x) = f_Y(y)$ for every (x, y) with $f_X(x) > 0$;
- (iii) $f_{X|Y}(x | y) = f_X(x)$ for every (x, y) with $f_Y(y) > 0$.

Example 5.3 (Reading conditional probabilities off a table). From the joint p.m.f. in Example 2.3, the conditional distribution of Y given $X = 1$ is obtained by dividing the row $X = 1$ by its row total $f_X(1) = 0.50$:

$$f_{Y|X}(y | 1) = \frac{(0.05, 0.20, 0.20, 0.05, 0)}{0.50} = (0.10, 0.40, 0.40, 0.10, 0).$$

This row is itself a probability distribution: it tells us how the express line behaves *when* the regular line has one customer.

Example 5.4 (Continuous conditioning). Let $f_{XY}(x, y) = 2$ on the triangle $0 \leq x \leq y \leq 1$, zero elsewhere. Then $f_X(x) = \int_x^1 2 dy = 2(1 - x)$ for $x \in (0, 1)$, so for any such x

$$f_{Y|X}(y | x) = \frac{2}{2(1 - x)} = \frac{1}{1 - x}, \quad x < y < 1.$$

Conditional on $X = x$, Y is uniform on $(x, 1)$. The unconditional distribution of Y is different: $f_Y(y) = \int_0^y 2 dx = 2y$ on $(0, 1)$. Since $f_{Y|X}(y | x) \neq f_Y(y)$ in general, X and Y are dependent — consistent with the non-rectangular support.

Remark 5.5 (Why we care for econometrics). The function $x \mapsto f_{Y|X}(\cdot | x)$ is the essential object of regression analysis. Its mean, $\mu_{Y|X}(x) = \int y f_{Y|X}(y | x) dy$, is the conditional expectation function — the population analogue of an OLS regression. We will study it in Unit 6 and use it as the structural object of regression in Units 9–10.

6 Several random variables

The bivariate framework extends to any finite collection $X_1, \dots, X_n$. The joint p.m.f. / p.d.f. is $f_{X_1 \dots X_n}(x_1, \dots, x_n)$ and the marginals are obtained by summing/integrating out unwanted coordinates. The variables are *mutually independent* if $f_{X_1 \dots X_n}(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$, which strengthens pairwise independence (recall Unit 1). An i.i.d. sample $X_1, \dots, X_n$ from a population with density f is the leading example in statistics, and provides the data foundation of the entire course from Unit 8 onwards.

Summary

- Joint p.m.f./p.d.f. describes (X, Y) on $\mathbb{R}^2$; $\mathbb{P}((X, Y) \in A)$ is a sum or double integral.
- *Marginals* are obtained by summing/integrating out the other variable. Marginals do not determine the joint distribution.
- *Independence* $\Leftrightarrow f_{XY}(x, y) = f_X(x)f_Y(y)$ with rectangular support.
- *Conditional distribution* $f_{Y|X}(y | x) = f_{XY}(x, y)/f_X(x)$ is a valid p.m.f./p.d.f. in y for every x with $f_X(x) > 0$.
- Independence $\Leftrightarrow f_{Y|X}(y | x) = f_Y(y)$ for all (x, y) .

Suggested exercises

1. Using the joint p.m.f. of Example 2.3, compute (a) $\mathbb{P}(X \geq 2, Y \geq 2)$, (b) $\mathbb{P}(|X - Y| \leq 1)$, (c) the conditional distribution of X given $Y = 2$.
2. Let $f_{XY}(x, y) = c(x + y)$ on $[0, 1]^2$ and zero elsewhere. Find c , the marginals, and the conditional distribution $f_{Y|X}(y | x)$. Are X and Y independent?
3. (Geometric probability.) A point (X, Y) is uniformly distributed on the unit disc. Find the marginal density of X and the conditional density of Y given $X = x$.
4. Suppose $f_{XY}(x, y) = ce^{-x-2y}$ for $x, y \geq 0$. Show that $X \perp Y$, find c and the marginals, and compute $\mathbb{P}(X < Y)$.
5. (Twin's IQ.) Let X and Y be the IQs of two identical twins, modelled jointly as bivariate normal with means 100, equal variances σ^2 , and correlation $\rho = 0.85$. Without doing the integrals, explain why $f_{Y|X}(\cdot | x)$ has mean $100 + 0.85(x - 100)$ and variance $(1 - 0.85^2)\sigma^2$. (Bivariate normal will be derived in Unit 7.)